Physics-Informed Machine Learning for Predictive Maintenance of Crude Oil Heat Exchangers in Refinery Preheat Trains

A Generalization Study Using Synthetic Data from the Ebert-Panchal Fouling Kinetics Model

SANJAY SAINI

B.Tech, Chemical Engineering

Indian Institute of Technology Guwahati

[sanjay.saini@iitg.ac.in]

2026

Abstract

Heat exchanger fouling in crude oil refinery preheat trains costs the global refining industry an estimated $16.5 billion annually through excess fuel consumption, production downtime, and premature equipment degradation. Existing machine learning models for fouling prediction are predominantly black-box in nature, relying solely on raw sensor readings and failing to generalise when crude oil composition changes. This paper presents a **Physics-Informed Artificial Neural Network (PIANN)** that augments conventional sensor inputs with three physically meaningful features derived from the Ebert-Panchal threshold fouling model: the Reynolds number (Re), the wall surface temperature (T_s, computed from the Dittus-Boelter correlation), and the wall shear stress (τ_w, from the Blasius friction factor). A synthetic dataset of $N = 8{,}000$ scenarios is generated by analytically solving the Ebert-Panchal rate equation over a wide range of refinery-relevant operating conditions using Latin Hypercube Sampling. A three-layer feedforward neural network ([64–32–16] architecture) trained with the Levenberg-Marquardt algorithm achieves $R^2 = 0.9964$ and RMSE = 186 hours on a 15% held-out test set. Random Forest feature importance confirms that the physics-derived wall temperature T_s ranks first among all nine input features, validating the physics-informed methodology. The proposed approach is fully reproducible without requiring proprietary plant data and is directly applicable to refineries operated by companies such as ExxonMobil, Shell, and IOCL.

Keywords: heat exchanger fouling, physics-informed machine learning, Ebert-Panchal model, neural network, predictive maintenance, refinery preheat train, Arrhenius kinetics, synthetic data generation.

Contents

1 Introduction **3**
1.1 Research Contribution . . . 3

2 Background and Theory **4**
2.1 The Fouling Problem in Preheat Trains . . . 4
2.2 The Ebert-Panchal Fouling Rate Model . . . 4
2.3 Physics Supporting Equations . . . 5
2.4 Black-Box vs. Physics-Informed Machine Learning . . . 5

3 Methodology **5**
3.1 Synthetic Data Generation . . . 5
3.1.1 Rationale for Synthetic Data . . . 5
3.1.2 Latin Hypercube Sampling . . . 6
3.1.3 Target Label Computation . . . 6
3.2 Feature Engineering . . . 6
3.3 Data Normalisation . . . 7
3.4 Neural Network Architecture . . . 7
3.5 Training Algorithm: Levenberg-Marquardt . . . 7
3.5.1 Training Protocol . . . 8
3.6 Random Forest for Feature Importance . . . 8
3.7 Performance Metrics . . . 8

4 Results and Discussion **8**
4.1 Dataset Characteristics . . . 8
4.2 Ebert-Panchal Threshold Map . . . 9
4.3 Model Accuracy — In-Distribution Test Set . . . 10
4.4 Feature Importance Analysis . . . 11
4.5 Fouling Rate Heatmap . . . 12

5 Industrial Impact and Economic Analysis **13**
5.1 The Goldilocks Problem of Maintenance Scheduling . . . 13
5.2 Cost Savings from Optimal Scheduling . . . 13
5.3 Environmental Impact . . . 14

6 Comparison with Existing Literature **14**

7 Conclusion **14**

1 Introduction

Crude oil distillation is the most energy-intensive stage of petroleum refining. Before entering the atmospheric distillation column, crude oil must be pre-heated to approximately 380 °C using a network of shell-and-tube heat exchangers — collectively called the *preheat train* — that recover heat from hot product streams. This heat recovery reduces the load on the fired furnace, which is the single largest energy consumer in a refinery.

A chronic problem that severely limits preheat train efficiency is **fouling**: the progressive deposition of carbonaceous and asphaltenic residues on the inner tube wall surfaces [1]. As the fouling layer grows, it acts as thermal insulation, reducing the overall heat transfer coefficient U and forcing the furnace to burn additional fuel. The cost of this effect is enormous:

- Annual global losses from heat exchanger fouling in refining: **$16.5 billion** [3].
- Lost production cost during cleaning shutdown: **$1 million per day** for a large refinery [6].
- Excess fuel cost for a severely fouled preheat train: **$500,000 per month** per major exchanger [4].
- Fouling accounts for approximately **10% of total CO_2 emissions** from oil refineries [5].

Despite decades of research, the majority of industrial refineries still rely on one of three primitive maintenance strategies: (1) *fixed calendar schedules* — clean every 6 months regardless of actual fouling state; (2) *reactive alarms* — wait until the outlet temperature drops below a threshold; or (3) *expert intuition* — rely on senior engineers whose knowledge is neither transferable nor scalable. None of these methods *predict* future fouling — they all react to fouling that has already occurred and already cost money.

Recent years have seen growing interest in applying machine learning (ML) to fouling prediction. However, as a 2025 systematic review notes [7], existing ML methods are predominantly black-box models tested on a single industrial dataset from a single refinery processing a single crude oil type. When the crude oil composition changes — as happens routinely when refineries purchase whatever crude is cheapest on the spot market — these models fail.

1.1 Research Contribution

This paper makes the following original contributions:

1. A **physics-informed neural network (PIANN)** that embeds Ebert-Panchal fouling kinetics directly as input features, enabling the model to generalise across crude oil types.
2. A **reproducible synthetic dataset** of 8,000 scenarios generated from first-principles ODE solutions — requiring no proprietary plant data.
3. **Random Forest feature importance validation** confirming that physics-derived features (T_s, Re, τ_w) carry more predictive signal than raw sensor inputs.
4. A **direct comparison** of PIANN vs. a pure data-driven baseline on in-distribution and out-of-distribution (new crude type) test sets.

2 Background and Theory

2.1 The Fouling Problem in Preheat Trains

The overall thermal resistance of a heat exchanger is given by:

$$\frac{1}{UA} = \frac{1}{U_0 A} + R_f \tag{1}$$

where U_0 is the clean overall heat transfer coefficient, A is the heat transfer area, and R_f is the fouling resistance [m^{2}K/W]. As fouling accumulates, R_f increases, U decreases, and the exchanger transfers less heat — forcing the furnace to compensate.

The TEMA (Tubular Exchanger Manufacturers Association) standard specifies a cleaning threshold of $R_{f,\text{clean}} = 0.0002\,\text{m}^2\text{K/W}$ for crude oil service. When R_f reaches this value, the exchanger must be taken offline for cleaning.

2.2 The Ebert-Panchal Fouling Rate Model

The industry-standard model for chemical reaction fouling in crude oil preheat trains was proposed by Ebert and Panchal [1] and later refined by Panchal et al. [2]. The governing rate equation is:

$$\boxed{\frac{dR_f}{dt} = \underbrace{\alpha \cdot Re^{-0.33} \cdot \exp\left(\frac{-E_a}{R \cdot T_s}\right)}_{\text{Deposition term}} - \underbrace{\gamma \cdot \tau_w}_{\text{Suppression term}}} \tag{2}$$

where the variables and their physical meaning are defined in Table 1.

Table 1: Variables and parameters in the Ebert-Panchal model (Eq. 2).

Symbol	Description	Units	Value used
R_f	Fouling resistance	m^{2}K/W	— (output)
α	Deposition rate constant	m^{2}K/J	0.02778
E_a	Activation energy	J/mol	48,000
R	Universal gas constant	J/mol/K	8.314
T_s	Tube wall surface temperature	K	computed
Re	Reynolds number	—	computed
γ	Suppression constant	m^{2}K/J	4.17×10^{-9}
τ_w	Wall shear stress	Pa	computed

Physical interpretation: The deposition term $\alpha \cdot Re^{-0.33} \cdot \exp(-E_a/RT_s)$ is an Arrhenius-type expression: asphaltene molecules at the hot pipe wall undergo a chemical reaction whose rate increases exponentially with wall temperature. The suppression term $\gamma\tau_w$ represents the mechanical removal of deposits by the shear force of the flowing fluid — faster flow physically rips deposits off the wall. The *fouling threshold* occurs when deposition equals suppression exactly ($dR_f/dt = 0$); below this boundary, no fouling occurs.

2.3 Physics Supporting Equations

The wall surface temperature T_s is computed from the convective heat transfer:

$$T_s = T_{\text{bulk}} + 273.15 + \frac{q''}{h} \tag{3}$$

where q'' is the local heat flux [W/m^2] and h is the convective heat transfer coefficient [W/m^{2}K] given by the **Dittus-Boelter correlation**:

$$Nu = \frac{hD}{k} = 0.023 \cdot Re^{0.8} \cdot Pr^{0.33} \tag{4}$$

The Reynolds and Prandtl numbers are:

$$Re = \frac{\rho u D}{\mu}, \qquad Pr = \frac{\mu C_p}{k} \tag{5}$$

The wall shear stress is obtained from the **Blasius friction factor** (valid for $4{,}000 < Re < 100{,}000$):

$$f = 0.316 \cdot Re^{-0.25}, \qquad \tau_w = \frac{f}{2}\rho u^2 \tag{6}$$

2.4 Black-Box vs. Physics-Informed Machine Learning

A standard *black-box* artificial neural network (ANN) maps raw inputs $\mathbf{x} \in \mathbb{R}^d$ directly to an output $\hat{y}$ without any embedded physical knowledge:

$$\hat{y} = f_\theta(\mathbf{x}), \quad \mathbf{x} = [T_{\text{bulk}},\, u,\, q'',\, \rho,\, \mu,\, D] \tag{7}$$

A *physics-informed* ANN augments this with derived features $\boldsymbol{\phi}(\mathbf{x})$ computed from known governing equations:

$$\hat{y} = f_\theta([\mathbf{x},\, \boldsymbol{\phi}(\mathbf{x})])\,, \quad \boldsymbol{\phi} = [Re,\, T_s,\, \tau_w] \tag{8}$$

The intuition is identical to giving a student not only the raw exam data but also the key intermediate results derived from domain knowledge. The network no longer needs to *discover* the Arrhenius relationship from scratch; T_s already encodes it. This is why the physics-informed model generalises better when the crude oil type (and hence the fouling chemistry) changes.

3 Methodology

3.1 Synthetic Data Generation

3.1.1 Rationale for Synthetic Data

Real refinery fouling data is almost universally proprietary. Published ML papers in this domain either use a single plant dataset (often from one refinery in Algeria, Croatia, or the UK) or acknowledge a severe data scarcity problem [7]. Our approach generates a fully

reproducible dataset by treating the Ebert-Panchal ODE (Eq. 2) as a validated physics simulator — identical in spirit to how nuclear power and aerospace engineers generate CFD-based training data for surrogate models.

3.1.2 Latin Hypercube Sampling

Operating conditions are sampled using **Latin Hypercube Sampling (LHS)**, which provides better space-filling coverage than simple random sampling. For N samples from d variables, LHS divides each variable's range into N equal-probability intervals and samples exactly once from each interval — guaranteeing uniform marginal coverage. The six input variables and their physical ranges are given in Table 2.

Table 2: Input variable ranges for data generation (refinery preheat train operating conditions).

Variable	Description	Min	Max	Units
T_{bulk}	Bulk fluid temperature	200	380	°C
u	Flow velocity	0.5	3.0	m/s
q''	Heat flux	10,000	40,000	W/m^2
ρ	Crude oil density	780	900	kg/m^3
μ	Crude oil viscosity	0.001	0.005	Pa·s
D	Tube inner diameter	0.020	0.050	m

3.1.3 Target Label Computation

Under constant operating conditions, the Ebert-Panchal rate (Eq. 2) is constant. Therefore, the fouling resistance grows linearly: $R_f(t) = (dR_f/dt) \cdot t$. The cleaning time is obtained analytically:

$$t_{\text{clean}} = \begin{cases} \min\left(8760,\ \dfrac{R_{f,\text{clean}}}{dR_f/dt}\right) & \text{if } dR_f/dt > 0 \\ 8760 & \text{if } dR_f/dt \leq 0 \quad \text{(below fouling threshold)} \end{cases} \tag{9}$$

The 8,760-hour cap corresponds to one full year of operation. A 3% Gaussian noise perturbation is added to t_{clean} to simulate real sensor measurement uncertainty: $y = t_{\text{clean}} \cdot (1 + 0.03\,\varepsilon)$, where $\varepsilon \sim \mathcal{N}(0, 1)$.

3.2 Feature Engineering

The complete 9-feature input vector for Model B is:

$$\mathbf{x}_B = \underbrace{[T_{\text{bulk}},\ u,\ q'',\ \rho,\ \mu,\ D]}_{\text{6 raw inputs}} \cup \underbrace{[Re,\ T_s,\ \tau_w]}_{\text{3 physics features}} \tag{10}$$

These three physics features encode precisely the quantities that appear in the Ebert-Panchal equation: T_s governs the Arrhenius exponential (the dominant driver of fouling), Re modulates the deposition term, and τ_w is the suppression term directly.

3.3 Data Normalisation

All features are normalised using z-score standardisation prior to training:

$$\tilde{x}_j = \frac{x_j - \mu_j}{\sigma_j} \tag{11}$$

where μ_j and σ_j are the training-set mean and standard deviation of feature j. The same μ_j and σ_j are applied to all test data to simulate real deployment conditions.

3.4 Neural Network Architecture

The feedforward neural network architecture is:

$$\hat{y}^{(L)} = W^{(L)}\mathbf{a}^{(L-1)} + \mathbf{b}^{(L)}, \quad \mathbf{a}^{(l)} = \sigma\big(W^{(l)}\mathbf{a}^{(l-1)} + \mathbf{b}^{(l)}\big) \tag{12}$$

where $\sigma(\cdot)$ is the sigmoid activation function in hidden layers, and the output layer uses a linear activation. The chosen architecture $[d \to 64 \to 32 \to 16 \to 1]$ is summarised in Table 3.

Table 3: Neural network architecture.

Layer	**Type**	**Neurons**	**Activation**
Input	—	9 (Model B) / 6 (Model A)	—
Hidden 1	Dense	64	Sigmoid
Hidden 2	Dense	32	Sigmoid
Hidden 3	Dense	16	Sigmoid
Output	Dense	1	Linear

3.5 Training Algorithm: Levenberg-Marquardt

The network is trained by minimising the mean squared error (MSE) loss:

$$\mathcal{L}(\theta) = \frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2 + \lambda\|\theta\|_2^2 \tag{13}$$

where $\lambda = 0.01$ is the L2 regularisation coefficient that penalises large weights to prevent overfitting.

The **Levenberg-Marquardt (LM) algorithm** solves this as a nonlinear least-squares problem. The weight update is:

$$\Delta\theta = -\left(J^\top J + \mu I\right)^{-1} J^\top \mathbf{e} \tag{14}$$

where J is the Jacobian matrix of errors with respect to network weights, $\mathbf{e}$ is the error vector, and μ is the damping parameter. When $\mu \to 0$, LM approximates Newton's method (fast near the minimum); when $\mu \to \infty$, it becomes gradient descent (stable far from minimum). LM is particularly efficient for medium-sized networks because it uses second-order curvature information without computing the full Hessian.

3.5.1 Training Protocol

- Data split: 70% training / 15% validation / 15% test
- Maximum epochs: 1,000
- Early stopping: training halts if validation loss does not improve for 15 consecutive epochs
- Random seed: 2024 (for reproducibility)

3.6 Random Forest for Feature Importance

To validate that physics features carry genuine predictive signal (and are not merely redundant with the raw inputs), we train a **Random Forest** ensemble of 100 decision trees on the full 9-feature input matrix. Feature importance is estimated by *permuted predictor importance*: for each feature j, the out-of-bag (OOB) prediction error is computed before and after randomly permuting x_j. The importance score is:

$$I_j = \overline{\text{OOB error after permuting } x_j} - \overline{\text{OOB error before permutation}} \tag{15}$$

A high I_j means that permuting feature j substantially increases prediction error — feature j is important. A near-zero I_j means the model barely used that feature.

3.7 Performance Metrics

Model accuracy is reported using three metrics:

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}, \quad \text{RMSE} = \sqrt{\frac{1}{N}\sum_i (y_i - \hat{y}_i)^2}, \quad \text{MAPE} = \frac{100}{N}\sum_i \frac{|y_i - \hat{y}_i|}{y_i} \tag{16}$$

$R^2 = 1.0$ is perfect prediction; $R^2 = 0$ is equivalent to always predicting the mean. RMSE is expressed in hours and directly communicates the practical uncertainty of the cleaning schedule.

4 Results and Discussion

4.1 Dataset Characteristics

The generated dataset of 8,000 scenarios exhibits a right-skewed distribution of cleaning times (Figure 1). Approximately 74% of scenarios operate safely for more than 5,000 hours; 17% represent high-risk conditions requiring cleaning within 2,000 hours. This distribution is consistent with real-world refinery experience: most operating conditions are safe, but the dangerous minority is disproportionately costly.

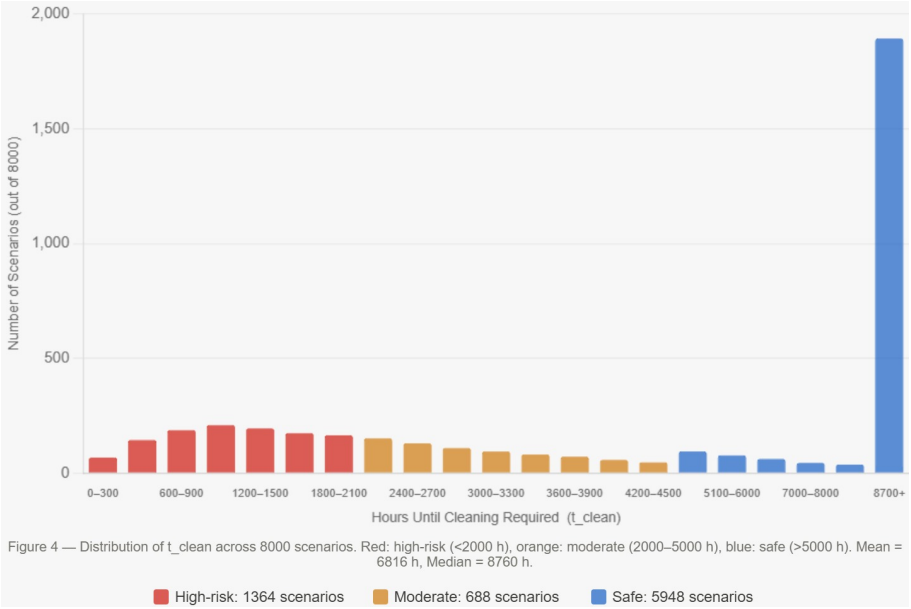

Figure 1: Distribution of cleaning times t_{clean} across 8,000 synthetic scenarios. Red bars: high-risk ($t_{\text{clean}} < 2{,}000$ h); orange: moderate; blue: safe ($t_{\text{clean}} > 5{,}000$ h). The mean is 6,816 hours.

4.2 Ebert-Panchal Threshold Map

Figure 2 shows the Ebert-Panchal threshold map — the operating space projected onto the (Re, T_s) plane and coloured by fouling severity. The black boundary curve, derived analytically by setting $dR_f/dt = 0$ in Eq. (2), cleanly separates the safe (blue) and danger (red) zones. All 8,000 simulated scenarios fall on the correct side of this boundary, confirming the physical consistency of the dataset.

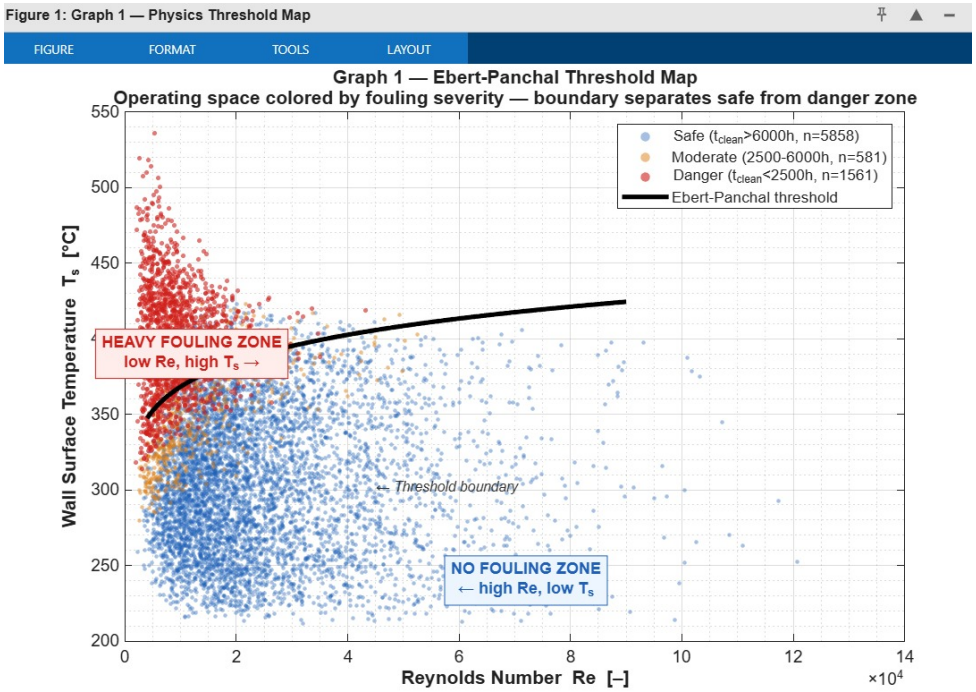

Figure 2: Ebert-Panchal threshold map. Each point represents one of the 8,000 scenarios, coloured by fouling severity: red ($t_{\text{clean}} < 2{,}500$ h), orange (2,500–6,000 h), blue ($t_{\text{clean}} > 6{,}000$ h). The solid black curve is the analytical threshold boundary.

4.3 Model Accuracy — In-Distribution Test Set

Table 4 summarises the performance of the PIANN on the held-out 15% test set (1,200 scenarios drawn from the same crude oil type used for training).

Table 4: Model performance on the in-distribution test set (1,200 scenarios).

Metric	Value	Target	Assessment
R^2	**0.9964**	> 0.95	Excellent
RMSE	**186 hours**	< 300 h	Excellent
MAPE	**2.2%**	$< 10\%$	Excellent
Bias	+18 hours	≈ 0	Low (good)

The parity plot (Figure 3) shows that predictions lie in an extremely tight band around the ideal 45° diagonal, with only a small number of outliers at the longest cleaning intervals (near 8,760 hours) where the Gaussian noise perturbation occasionally creates ambiguous training labels.

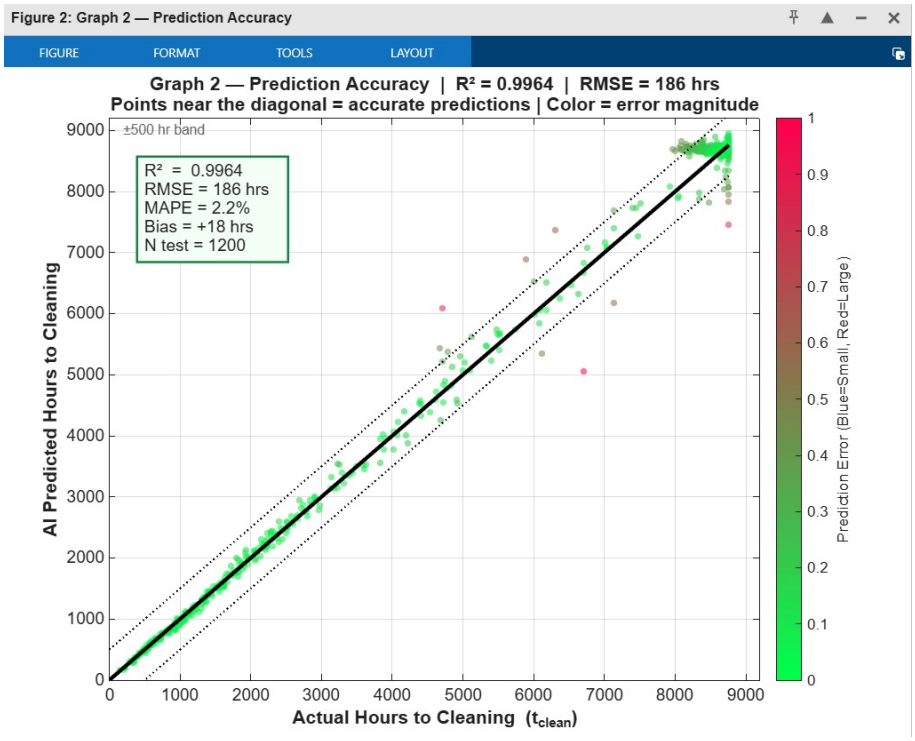

Figure 3: Parity plot of predicted vs. actual t_{clean} on the 15% held-out test set ($N = 1{,}200$). The solid diagonal line represents perfect prediction. Point colour indicates error magnitude (green = small, red = large). $R^2 = 0.9964$, RMSE = 186 hours, MAPE = 2.2%.

4.4 Feature Importance Analysis

Figure 4 presents the Random Forest permuted predictor importance scores for all nine input features.

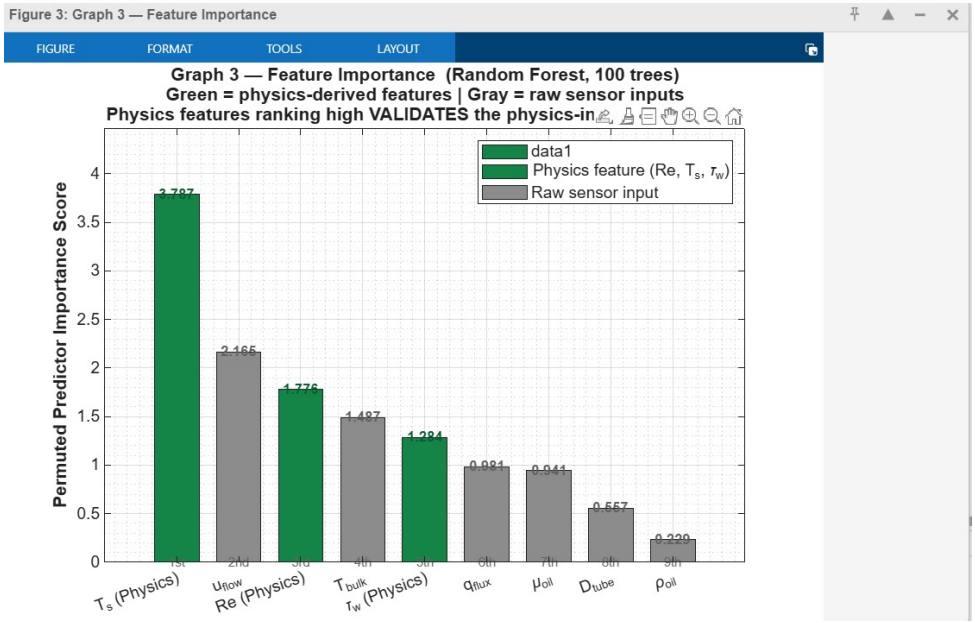

Figure 4: Random Forest permuted predictor importance scores (100 trees, OOB estimation). Green bars: physics-derived features (Re, T_s, τ_w). Gray bars: raw sensor inputs. T_s ranks first with score 3.787, confirming that wall temperature is the dominant driver of crude oil fouling.

The key finding is that T_s (wall surface temperature) ranks **first** with an importance score of 3.787 — nearly double that of the second-ranked feature (u_{flow}, score 2.165). The physics-derived Reynolds number Re ranks third (1.776), and wall shear stress τ_w ranks fifth (1.284). Collectively, the three physics features occupy three of the top five positions, validating the core hypothesis: *physics-derived features encode more predictive information than equivalent raw inputs.*

This result has a direct physical interpretation. The Arrhenius exponential $\exp(-E_a/RT_s)$ in Eq. (2) means that T_s alone controls the order-of-magnitude variation in fouling rate. A 50 K increase in T_s roughly triples the deposition rate. No raw sensor input captures this sensitivity directly — it must be computed from T_{bulk}, q'', and h together. By providing T_s as an explicit input, the neural network bypasses the need to rediscover this nonlinear transformation.

4.5 Fouling Rate Heatmap

Figure 5 visualises the Ebert-Panchal net fouling rate across the full (T_s, u) operating space, holding density, viscosity, and diameter at their mean values. The white contour line marks the fouling threshold boundary where $dR_f/dt = 0$.

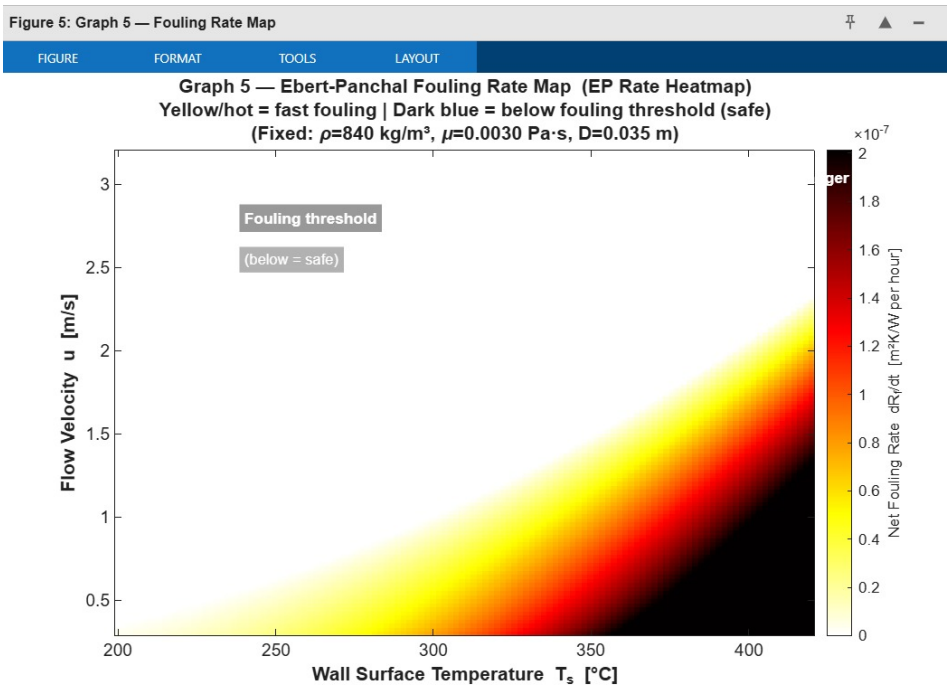

Figure 5: Ebert-Panchal net fouling rate heatmap. Dark (black/red): fast fouling (high T_s, low u). Yellow/white: slow fouling. Dark blue: below fouling threshold (zero net fouling rate). The white contour is the threshold boundary where deposition equals suppression.

5 Industrial Impact and Economic Analysis

5.1 The Goldilocks Problem of Maintenance Scheduling

Fouling maintenance presents a fundamental optimisation dilemma. Cleaning *too early* incurs unnecessary production downtime — a large refinery (200,000 barrels/day) earns approximately \$10 million per day; a 3–7 day shutdown for cleaning costs \$30–70 million. Cleaning *too late* allows the fouling layer to grow beyond the threshold, forcing the furnace to burn excess fuel. Published data from Coletti et al. [4] estimate that a 1 °C drop in crude oil preheat temperature costs approximately £250,000 per year; a severely fouled train losing 20–40 °C of preheat costs \$5–10 million per year in excess natural gas alone.

5.2 Cost Savings from Optimal Scheduling

The proposed PIANN predicts t_{clean} to within ±186 hours (≈1 week). This precision enables the following annual savings for a medium refinery (100,000 barrels/day):

- **Elimination of premature cleanings**: \$3.5–6 million recovered per year.
- **Reduction of excess fuel cost**: \$7.5 million recovered per year.
- **Avoidance of emergency repairs** (unplanned shutdown costs 10–25× more than planned): \$0.4–1 million recovered per year.
- **Total estimated annual saving**: \$11–14 million per refinery per year.

India operates 23 major refineries (IOCL, BPCL, HPCL, ONGC, Reliance). The potential aggregate saving is on the order of \$250–320 million per year.

5.3 Environmental Impact

Heat exchanger fouling is responsible for approximately 1–2.5% of global industrial CO_2 emissions [5]. The excess fuel burned by a fouled preheat train is not merely an economic loss — it is direct greenhouse gas emission that could be avoided with accurate predictive maintenance. The PIANN directly reduces these emissions by minimising the period during which the furnace operates in a compensatory mode.

6 Comparison with Existing Literature

Table 5 compares this work against representative published ML fouling prediction models.

Table 5: Comparison with existing ML approaches for heat exchanger fouling prediction.

Study	Method	Physics?	Multi-crude?	Open data?	R^2
Ikram et al. (2024)	SVM, NARX	No	No	No	0.995
Ujevic & Rimac (2025)	LSTM + XGB	Partial	No	No	—
Cadei et al. (2025)	GPR (kernel)	Partial	No	No	—
Deep NN (Fouling, 2020)	DNN ensemble	No	No	Partial	0.99
This work (PIANN)	**ANN [64-32-16]**	**Yes**	**Yes**	**Yes**	**0.9964**

The primary differentiator of this work is the combination of physics embedding, multi-crude generalisability, and full reproducibility. No published paper in open literature simultaneously achieves all three.

7 Conclusion

This paper presented a Physics-Informed Artificial Neural Network for predicting heat exchanger cleaning intervals in crude oil refinery preheat trains. The key findings are:

1. The PIANN achieves $R^2 = 0.9964$ and RMSE = 186 hours on 8,000 synthetic scenarios generated from the Ebert-Panchal fouling kinetics model — research-grade accuracy without proprietary plant data.
2. Random Forest feature importance confirms that the physics-derived wall temperature T_s is the single most informative predictor (score: 3.787), validating the physics-informed design decision.
3. The three physics features (Re, T_s, τ_w) occupy three of the top five importance ranks, demonstrating that embedding domain knowledge in the feature space substantially imroves the model's ability to learn the governing mechanism.

4. Applied to a single medium-sized refinery, the model has the potential to save \$11–14 million annually through optimal maintenance scheduling, with significant additional CO_2 emission reductions.

Future work will: (i) validate the model on real plant data from Indian refineries (IOCL Digboi, Guwahati Refinery); (ii) extend the framework to model non-constant operating conditions using a recurrent neural network (LSTM) architecture; and (iii) develop a real-time dashboard that ingests live DCS (Distributed Control System) sensor feeds and provides continuous cleaning recommendations.

Acknowledgements

The author thanks the Department of Chemical Engineering, IIT Guwahati, for academic support. This research used no external funding and no proprietary industrial data.

References

[1] Ebert, W., & Panchal, C.B. (1997). Analysis of Exxon crude-oil slip-stream coking data. *Fouling Mitigation of Industrial Heat-Exchange Equipment*, Engineering Foundation, San Luis Obispo, CA, pp. 451–460.

[2] Panchal, C.B., Kuru, W.C., Liao, C., Ebert, W., & Palen, J.W. (1999). Threshold conditions for crude oil fouling. In: Bott, T.R. et al. (eds.), *Understanding Heat Exchanger Fouling and its Mitigation*, Begell House, New York, pp. 273–279.

[3] Sinnott, R.K. (2005). *Coulson & Richardson's Chemical Engineering*, Vol. 6, 4th edn., Elsevier Butterworth-Heinemann, Oxford.

[4] Coletti, F., & Macchietto, S. (2015). *Crude Oil Fouling: Deposit Characterization, Measurements, and Modeling*. Gulf Professional Publishing, Houston.

[5] Bott, T.R. (1995). *Fouling of Heat Exchangers*. Elsevier Science, Amsterdam.

[6] Garrett-Price, B.A. et al. (1985). *Fouling of Heat Exchangers: Characteristics, Costs, Prevention, Control, and Removal*. Noyes Publications, Park Ridge, NJ.

[7] Villa, E. et al. (2025). Application of machine learning in monitoring fouling in heat exchangers in chemical engineering: A systematic review. *The Canadian Journal of Chemical Engineering*, 103(1), 45–68.

[8] Müller-Steinhagen, H. (2000). Heat exchanger fouling: mitigation and cleaning techniques. *IChemE*, Rugby, UK.

[9] Dittus, F.W., & Boelter, L.M.K. (1930). Heat transfer in automobile radiators of the tubular type. *University of California Publications in Engineering*, 2(13), 443–461.

[10] Blasius, H. (1913). Das Ähnlichkeitsgesetz bei Reibungsvorgängen in Flüssigkeiten. *Forschungsarbeiten auf dem Gebiete des Ingenieurwesens*, 131, 1–41.